MATH 415/515 Advanced Linear Algebra
Exam 2 Review

1 List of Topics

1. Exam 1 material
   (a) Review
      i. Vector space, V
      ii. Subspace, W
      iii. Linear combination, span, spanning set
      iv. Linear independence
      v. Basis
      vi. Dimension
      vii. Eigenvalue, eigenvector, eigenspace, algebraic multiplicity, geometric multiplicity
      viii. Null space, $\mathrm{NS}(A)$, column space, $\mathrm{CS}(A)$, row space, $\mathrm{RS}(A)$
   (b) Inner product on $\mathbb{R}^n$, dot product
      i. Length, norm, $\|\mathbf{v}\|$
      ii. Unit vector, $\hat{\mathbf{v}}$
      iii. Distance, $\|\mathbf{v} - \mathbf{u}\|$
      iv. Angle, θ
      v. Orthogonality
         A. Orthogonal complement, $W^\perp$
         B. Orthogonal set, orthogonal basis
         C. Orthogonal projection of $\mathbf{x}$ onto a subspace W, $\mathbf{x}' = \mathrm{proj}_W(\mathbf{x})$
         D. The component of $\mathbf{x}$ orthogonal to W, $\mathbf{z}$
         E. Orthonormal set, orthonormal basis
         F. Orthogonal matrix, U
         G. The Gram-Schmidt process
         H. QR-factorization
         I. Least-squares solution to $A\mathbf{x} = \mathbf{b}$, normal equations
2. Curve fitting
   (a) Design matrix, X, parameter vector, $\boldsymbol{\beta}$, observation vector, $\mathbf{y}$
   (b) Residual vector, $\boldsymbol{\varepsilon}$, the linear regression model
   (c) Coefficient of determination, R^2
3. Inner product spaces
4. Fourier series
5. Orthogonal polynomials
   (a) Inner product, orthogonality
   (b) Location of roots
   (c) Three-term recursion formula
6. Applications to linear differential equations

(a) Linear transformation, kernel, eigenvalue, eigenfunction
(b) Sturm-Liouville problems
(c) Bessel functions
(d) Orthogonal polynomials: Legendre, Chebyshev, Hermite

7. [MATH 515] QR algorithm

2 Sample Questions

Short-answer questions for Exam 2 may come from the following list. If I think of additional problems, I reserve the right to include them on Exam 2.

Review (Exam 1)

1. Define a subspace W of a vector space V.
2. Define a linear combination of the vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$.
3. Define Span $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$.
4. Define linear independence for $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$.
5. Define a basis of a subspace W.
6. Define the dimension of a subspace W. Include definitions for both a finite-dimensional subspace and an infinite-dimensional subspace.
7. State the Basis Theorem.
8. Define an eigenvalue, λ, and an associated eigenvector, $\mathbf{v}$, of a matrix A.
9. State a theorem relating the concepts of eigenvectors and linear independence.
10. State the definition of a linear transformation, $T : V \to W$.

Inner product (Exam 1)

11. Define the inner product on $\mathbb{R}^n$.
12. Define orthogonality between two vectors.

Orthogonal sets (Exam 1)

13. Define an orthogonal basis for a subspace W.
14. Define the orthogonal projection of $\mathbf{x}$ onto a subspace W, $\text{proj}_W(\mathbf{x})$.
15. Define an orthonormal basis for a subspace W.

Orthogonal projections (Exam 1)

16. State the Orthogonal Decomposition Theorem.
17. State the Best Approximation Theorem.

The Gram-Schmidt process (Exam 1)

18. Explain the purpose of the Gram-Schmidt process. (What do we need before we can start the process and what does it produce?)
19. How does the Gram-Schmidt process work? (What happens at each stage of the process? A picture might help.)

Least-squares problems (Exam 1)

20. Define a least-squares solution of $A\mathbf{x} = \mathbf{b}$.
21. Define the normal equations for $A\mathbf{x} = \mathbf{b}$.
22. How do we calculate the error associated with a least-squares solution?

Curve fitting

23. State the linear regression model, and label all four pieces of the model.
24. Explain why even though a curve might not be linear, the regression model is still linear.
25. Be able to fit a curve to data.
26. Explain what the coefficient of determination, R^2 measures. What are the range of values, and what do those values mean?
27. What two cases must be considered when calculating the coefficient of determination, R^2?

Inner product spaces

28. Define a general inner product.
29. Define an inner product for $\mathbb{C}^n$.
30. Give two examples of an inner product that are not the usual inner product involving the dot product.
31. Be able to perform any of the eight calculations outlined as applications of an inner product for a given vector space and inner product.

Fourier series and orthogonal polynomials

32. What is a Fourier series? Why is it a topic in an advanced linear algebra course?
33. Be able to calculate a Fourier series with technology.
34. Define a sequence of orthogonal polynomials.
35. State a theorem that provides information about a sequence of orthogonal polynomials and its use as a basis and connection with the orthogonal complement.
36. Be able to use the three-term recursion formula for a sequence of orthogonal polynomials.
37. State a theorem concerning the roots of a sequence of orthogonal polynomials.

Applications for linear differential equations

38. How does solving a linear, homogeneous ODE relate to finding the kernel of a linear transformation?
39. How does solving a linear, homogeneous ODE relate to eigenvalues and eigenfunctions?
40. Define a Sturm-Liouville problem. Define a regular Sturm-Liouville problem.
41. Be able to use the μ-integrating factor method to put an ODE into Sturm-Liouville form and identify $w(x)$.
42. State a theorem that provides six facts about regular Sturm-Liouville problems.
43. Explain why Sturm-Liouville theory, which deals with ODEs, is being discussed in an advanced linear algebra course.
44. What are a Bessel functions? How are they similar and different from the families of orthogonal polynomials discussed, such as Legendre, Chebyshev, and Hermite polynomials?
45. Be able to take a Sturm-Liouville problem, find its inner product, its eigenvalues, and use Taylor series to find its associated eigenfunctions.
46. In the process for finding various polynomials, we assumed $y = \sum a_n x^n$. Walk through that process and explain every place where linear algebra is used.

QR Algorithm

47. Explain what the QR algorithm finds and how it works.

48. [MATH 515] Explain shifting in the QR algorithm.

49. [MATH 515] Explain deflating in the QR algorithm.

Technology note

50. Be prepared to use your laptop and some sort of computer algebra system, like Maple, MATLAB, Python, etc., to compute things like Fourier series, Gram-Schmidt with general inner products, finding the best approximation with general inner products, fitting data to curves, etc. You also need to have a plan for submitting those files and calculations to a D2L dropbox.

    If you have code from the homework that you would like to use, say for computing least squares or fitting data to a polynomial, you are free to use that. Just specify that you are doing so on the exam. Also, this means that if your code had any issues from the homework, now would be the time to fix that code.

    I recommend Maple or MATLAB, or a similar program, where you can put all of your computations in one file that can be submitted to D2L. Using online calculators may be fine for the homework, but you need to submit your calculations in a timely manner, which could be cumbersome.

3 Practice Problems

1. Prove that if an $n \times n$ matrix U satisfies $(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$, for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, then U is an orthogonal matrix. Hint: this holds for ANY vector in $\mathbb{R}^n$. Consider the standard basis $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ and consider what $U\mathbf{e}_i$ computes, for $i = 1, \ldots, n$.

2. Let U be an orthogonal matrix.

   (a) Prove that $\det(U) = \pm 1$. Hint: recall that $\det(A^T) = \det(A)$.

   (b) Prove that for any eigenvalue of U, $|\lambda| = \pm 1$. Hint: recall that $\|U\mathbf{x}\| = \|\mathbf{x}\|$.

3. A *Householder matrix* has the form $Q = I - 2\mathbf{u}\mathbf{u}^T$, where $\mathbf{u}$ is a unit vector. Prove that Q is an orthogonal matrix. Hint: compute $Q^T Q$.

4. Let X be the design matrix used to find the least-squares line to fit data $(x_1, y_1), \ldots (x_n, y_n)$. Use a theorem to prove that the normal equations have a unique solution if and only if there are at least two data points with different x-coordinates.

5. A certain experiment produces data $(1, 7.9), (2, 5.4), (3, -0.9)$. Find a least-squares fit of these points by a function of the form $y = A\cos(x) + B\sin(x)$. Show all of your work.

6. A healthy child's systolic blood pressure p (in millimeters of mercury) and weight w are approximately related by the equation $p = \beta_0 + \beta_1 \ln(w)$. Use the following experimental data to estimate the systolic blood pressure of a healthy child weighing 100 pounds. Show all of your work.

7. Consider the inner product for P_n where $\langle p, q\rangle = p(t_0)q(t_0) + \cdots + p(t_n)q(t_n)$ with $t = -1, 0, 1$ for P_2. Let $p(t) = 4 + t$ and $q(t) = 5 - 4t^2$.

   (a) Compute $\langle p, q\rangle$.

   (b) Compute $\|p\|$ and $\|q\|$.

   (c) Compute the orthogonal projection of q onto the subspace spanned by p.

8. Let $V = C[0, 1]$ with the inner product $\langle f, g\rangle = \int_0^1 f(t)g(t)\,dt$.

   (a) Compute $\langle f, g\rangle$, where $f(t) = 1 - 3t^2$ and $g(t) = t - t^3$.

   (b) Compute $\|f\|$.

9. Let $V = C[-1, 1]$ with the inner product $\langle f, g\rangle = \int_{-1}^1 f(t)g(t)\,dt$. Find an orthogonal basis for the subspace spanned by the polynomials 1, t, and t^2. This process will create the Legendre polynomials, since the inner product is the same inner product as that for the Legendre polynomials.

10. Let
$$\bar{x} = \frac{1}{n}(x_1 + \cdots + x_n)$$
and
$$\bar{y} = \frac{1}{n}(y_1 + \cdots + y_n).$$
Prove that the least-squares line for the data $(x_1, y_1), \ldots, (x_n, y_n)$ must pass through $(\bar{x}, \bar{y})$. That is, prove that $\bar{x}$ and $\bar{y}$ satisfy the linear equation $\bar{y} = \beta_0 + \beta_1\bar{x}$. Hint: derive this equation from the vector equation $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. Denote the first column of X by $\mathbb{1}$, which is the column vectors of all 1s. Use the fact that the residual vector $\boldsymbol{\varepsilon}$ is orthogonal to the columns of X and hence is orthogonal to $\mathbb{1}$.

11. Laguerre's equation is

$$x\frac{d^2L}{dx^2} + (1-x)\frac{dL}{dx} + \lambda L = 0,$$

with $0 < x < \infty$. We will derive the Laguerre polynomials, $L_n(x)$, which solves Laguerre's equation.

(a) Identify the inner product for the Laguerre polynomials. Note that the given ODE is not in Sturm-Liouville form, and so we cannot identify $w(x)$. Use the μ-integrating factor method from ODEs to collapse the first two terms into a single derivative, which will put the ODE in Sturm-Liouville form.

It can be shown that $\langle L_n(x), L_n(x)\rangle = 1$.

(b) Laguerre's equation is non-singular, and so we can assume a series solution of the form

$$L(x) = \sum_{n=0}^{\infty} a_n x^n.$$

Use the series methods illustrated in class and in the homework to find a recurrence relation for a_n. Specifically, you should find a recurrence relation of the form

$$a_{n+1} = g(a_n, \lambda),$$

where $g(a_n, \lambda)$ is a function of a_n and λ.

(c) We need bounded solutions, which means that the series solution must terminate in a finite number of terms. Use the recurrence relation from (a) and set $a_{n+1} = 0$, which will guarantee that the series terminates for some value of n. Solve for λ and show that $\lambda = n$.

(d) We can now derive formulas for Laguerre polynomials. First, we need to update the recurrence relation from (a) with $\lambda = n$. Note that n is now used to specify the order of Laguerre's equation, not the index for the recurrence. So rewrite the recurrence relation from (a) using the index k with $\lambda = n$. In this way $a_{k+1} = g(a_k, n)$.

Use the following procedure to find the Laguerre polynomials.

i. Pick a value of n, for $n = 0, 1, 2, \ldots$.

ii. Use k to generate a_{k+1} until $a_{k+1} = 0$, which is guaranteed by (b). Use $k = 1, 2, 3, \ldots$.

iii. Write out the polynomial $L_n(z)$ using the coefficients in (c), though you can reduced every coefficient to a function of a_0.

iv. We normalize Laguerre polynomials so that $a_0 = 1$. Use this to determine the final form of $L_n(x)$, which should not include any unknown coefficients.

Show that $L_0(x) = 1$, $L_1(x) = 1 - x$, $L_2(x) = 1 - 2x + \frac{1}{2}x^2$, and $L_3(x) = 1 - 3x + \frac{3}{2}x^2 - \frac{1}{6}x^3$.

It can be shown that, in general,

$$L_n(x) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{x^k}{k!},$$

though you do not need to do that for this problem. There is also an *associated Laguerre's equation*,

$$x\frac{d^2y}{dx^2} + (\alpha + 1 - x)\frac{dy}{dx} + nx = 0,$$

and the solutions are the *associated Laguerre's polynomials*, given by

$$L_n^{\alpha}(x) = \sum_{k=0}^{n} (-1)^k \binom{n+\beta}{n-k} \frac{x^k}{k!}.$$

In this case it can be shown that

$$\langle L_n^{\alpha}(x), L_n^{\alpha}(x)\rangle = \frac{\Gamma(n+\alpha+1)}{n!}.$$

12. Another common class of polynomials that appears frequently with PDEs is the *Jacobi polynomial*, $P_n^{(\alpha,\beta)}$, which generates the Legendre polynomials for $\alpha = 0$ and $\beta = 0$, and which generates the Chebyshev polynomials for $\alpha = -1/2$ and $\beta = -1/2$. *Jacobi's equation* is

$$\left(1 - x^2\right) P'' + [\beta - \alpha - (\alpha + \beta + 2)x] P' + \lambda P = 0.$$

(a) Identify the inner product for the Jacobi polynomials. Note that the given ODE is not in Sturm-Liouville form, and so we cannot identify $w(x)$. Use the μ-integrating factor method from ODEs to collapse the first two terms into a single derivative, which will put the ODE in Sturm-Liouville form.

I did this, just to see what I was asking you to do, and it's pretty crazy. I recommend you try, though, because if you can do this, then you can do anything I could put on the exam. Here are some hints. When you calculate the integral for x, split it into two integrals. One will use partial fractions, and the other will use a u-substitution. You should get

$$\ln|\mu| = \frac{\beta - \alpha}{2} \ln\left|\frac{1+x}{1-x}\right| + \frac{\alpha + \beta + 2}{2} \ln\left|1 - x^2\right|.$$

Then use rules of logarithms to show that $\mu = (1-x)^{\alpha+1}(1+x)^{\beta+1}$. Then it gets easier, and you should find $w(x) = (1-x)^{\alpha}(1+x)^{\beta}$.

(b) Jacobi's equation is non-singular, and so we can assume a series solution of the form

$$P^{(\alpha,\beta)}(x) = \sum_{n=0}^{\infty} a_n x^n.$$

(We do not need to use the Method of Frobenius.) Use the series methods illustrated in class and in the homework to find a recurrence relation for a_n. Specifically, you should find a recurrence relation of the form

$$a_{n+2} = g(a_n, a_{n+1}, \alpha, \beta, \lambda).$$

(c) We need finite solutions, which means that the series solution must terminate in a finite number of terms. Use the recurrence relation from (a) and set both $a_{n+2} = 0$ and $a_{n+1} = 0$, which will guarantee that the series terminates for some value of n. Solve for the λ that will produce finite solutions to Jacobi's equation.

(d) Personally, I've never taken this information and used it to find the general Jacobi polynomials, so you could try to find them for practice, but I don't know if that will be easy or hard. The normalization for the Jacobi polynomials is $P_n^{(\alpha,\beta)}(1) = \binom{n+\alpha}{n}$.

13. Given a function, $f(x)$, assume that you want to write $f(x)$ as a linear combination of any of the specialty functions we have discussed (Fourier sines and cosines, Bessel functions, Legendre polynomials, Chebyshev polynomials, Hermite polynomials, Laguerre polynomials, or Jacobi polynomials) so that

$$f(x) = \sum_{n=0}^{\infty} a_n p_n(x),$$

where $p_n(x)$ is the specialty function of interest, derive a formula for a_n using the principle of orthogonality. You should be able to come up with a formula for any of the available functions, though some will be nicer than others, since we know the norms for some of the functions.

Theorem (The Invertible Matrix Theorem)**.** Let A be an $n \times n$ matrix. The following are equivalent.

(1) A is an invertible matrix.

(2) A is row equivalent to the $n \times n$ identity matrix.

(3) A has n pivot rows.

(4) The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

(5) The columns of A form a linearly independent set.

(6) The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one/injective.

(7) The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b} \in \mathbb{R}^n$.

(8) The columns of A span $\mathbb{R}^n$.

(9) The linear transformation $x \mapsto A\mathbf{x}$ is onto/surjective.

(10) There is an $n \times n$ matrix C such that $CA = I$.

(11) There is an $n \times n$ matrix D such that $AD = I$.

(12) A^T is an invertible matrix.

(13) $\det(A) \neq 0$.

(14) The columns of A form a basis of $\mathbb{R}^n$.

(15) $\mathrm{CS}(A) = \mathbb{R}^n$.

(16) $\mathrm{rk}\,(A) = n$.

(17) $\nu\,(A) = 0$.

(18) $\mathrm{NS}(A) = \{\mathbf{0}\}$.

(19) The number 0 is not an eigenvalue of A.